

Math Primer

(2. Relations and Functions)

Contents



- ▶ Cartesian product of sets
- ▶ Relations
- ▶ Functions
- ▶ Named Functions
- ▶ Properties of Functions
- ▶ Composition
- ▶ Inverse
- ▶ Images and Pre-Images of Sets



Leibnitz
(1646-1716)

Cartesian product of Sets

- ▶ Given two sets P, Q , the cartesian product is all ordered pairs from set P and Q . In other words,

$$P \times Q = \{ (p, q) : p \in P, q \in Q \}$$

- ▶ If either P or Q is a null set, then $P \times Q = \phi$

- ▶ Example :

$$P = \{a, b\} \text{ and } Q = \{1, 2, 3\}, \text{ then } P \times Q = \{ (a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3) \}$$

- ▶ if $|P| = m$ and $|Q| = n$, then $|P \times Q| = mn$

- ▶ Two ordered pairs are equal iff the first elements are the same and the second elements are the same.

$$(a, 1) \neq (1, a)$$

Cartesian product of Sets

Exercise 1 : If $(x + 1, y - 2) = (3, 1)$, find the values of x and y .

$$x = 2, y = 3$$

Exercise 2 : If $P = \{a, b, c\}$ and $Q = \{r\}$, is $P \times Q \neq Q \times P$?

$$P \times Q = \{(a, r), (b, r), (c, r)\} \quad Q \times P = \{(r, a), (r, b), (r, c)\}$$

Exercise 3 : If $P = \{1, 2\}$, form the set $P \times P \times P$.

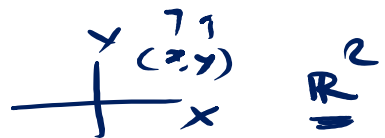
"ordered triplet"

$$P \times P = \{(1, 1), (1, 2), (2, 1), (2, 2)\} \quad P \times P \times P = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}$$

Exercise 4 : What do the cartesian products $\mathbb{R} \times \mathbb{R}$ and $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ represent?

$$\mathbb{R} \times \mathbb{R} \quad \mathbb{R} \times \mathbb{R} \times \mathbb{R}$$

2D space 3D space



Exercise 5 : $A = \{1, 2\}$ and $B = \{3, 4\}$. How many subsets will $A \times B$ have?

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\} \quad 2^4 = 16$$

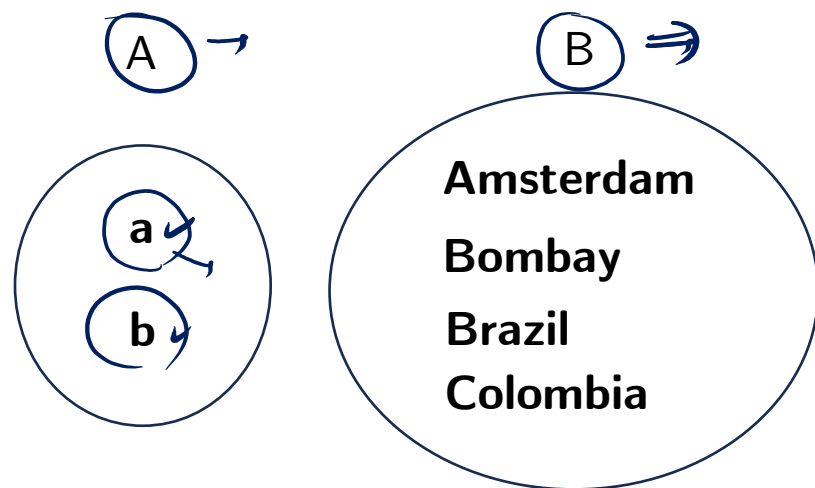
Exercise 6 : The Cartesian product $A \times A$ has 9 elements among which are found $(-1, 0)$ and $(0, 1)$. Find the set A and the remaining elements of $A \times A$.

$$A = \{-1, 0, 1\} \quad A \times A = \{(-1, -1), (-1, 0), (-1, 1), (0, -1), (0, 0), (0, 1), (1, -1), (1, 0), (1, 1)\}$$

Relations

$$\left\{ \overset{\text{first}}{\underset{\text{second}}{(a, b)}}, (c, d), (e, f), (g, h) \right\} \subseteq A \times B$$

- ▶ A relation R from a non-empty set A to a non-empty set B is a subset of the cartesian product $A \times B$
- ▶ The first element and the second element are related in some way.
- ▶ The second element is called as the image of the first element.
- ▶ The set of first elements of the ordered pairs in R is called the domain
- ▶ The set of second elements of the ordered pairs in R is called the range
- ▶ The whole set B is called the codomain (Note $\text{range} \subseteq \text{codomain}$)

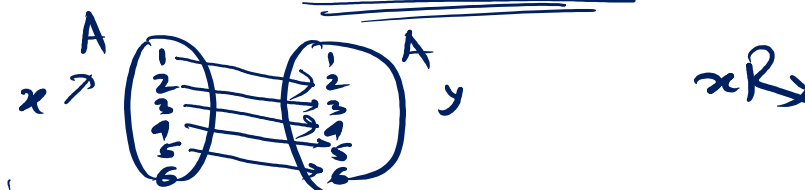


Example :

- ▶ $A \times B = \{ (a, \text{Amsterdam}), (a, \text{Bombay}), (a, \text{Brazil}), (a, \text{Colombia}), (b, \text{Amsterdam}), (b, \text{Bombay}), (b, \text{Brazil}), (b, \text{Colombia}) \}$
- ▶ Define a relation $R = \{ (x, y) : x \text{ is the first letter of } y, x \in A, y \in B \}$
- ▶ Then $R = \{ (a, \text{Amsterdam}), (b, \text{Bombay}), (b, \text{Brazil}) \}$
- ▶ Domain = $\{a, b\}$, Range = $\{\text{Amsterdam}, \text{Bombay}, \text{Brazil}\}$
- ▶ Codomain = $\{\text{Amsterdam}, \text{Bombay}, \text{Brazil}, \text{Colombia}\}$

Relations Representation

- ▶ A Relation can be represented in
 - ▶ Roster form ✓
 - ▶ Set builder form ✓
 - ▶ Arrow diagram ✓
- ▶ If (x, y) belongs to a Relation, we say x is related to y , denoted as $x \underline{\underline{R}} y$
- ▶ A Relation from $\underline{\underline{A}}$ to $\underline{\underline{A}}$ is also called as "a Relation on A"



Exercise 7:

Let $A = \{1, 2, 3, 4, 5, 6\}$. Define a relation R from A to A by $R = \{(x, y) : y = x + 1\} = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$

(i) Depict this relation using an arrow diagram. (ii) Write down the domain, codomain and range of R

set builder R
 $= (A) = \{2, 3, 4, 5, 6\}$

Exercise 8: How many Relations exist from P to Q if $|P| = m$ and $|Q| = n$

subset of $P \times Q$
 $|P \times Q| = mn$
 2^{mn} , ϕ , $\{ \}$

Properties of a Relation

- ▶ Let R be a Relation on a set A
- $R: A \rightarrow A$ $A = \{x\}$
- ▶ R is **Reflexive** iff $\forall x \in A, x R x$ $(x, x) \in R$
 - ▶ R is **Symmetric** iff $\forall x, y \in A, x R y \Rightarrow y R x$ $(x, y), (y, x)$
 - ▶ R is **Transitive** iff $\forall x, y, z \in A, x R y \text{ and } y R z \Rightarrow x R z$ $(x, y), (y, z), (x, z)$
 - ▶ R is an **Equivalence Relation** if it's Reflexive, Symmetric and Transitive

Exercise 9: Determine if this Relation is an equivalence relation. $A = \{1, 2, 3\}$, Relation on A is given as $R = \{(x, y) : x, y \in A, x \leq y\}$

$R = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

$x < y, y < x$ $x \leq y, y \leq x$ $x \leq x$

Exercise 10: Determine if this Relation is an equivalence relation. $C = \{0, 1\}$, Relation on C is given as $R = \{(x, y) : x, y \in C, x + y = \text{even}\}$

$R = \{(0, 0), (1, 1)\}$

odd + odd = even
even + even = even

Exercise 11: Determine if this Relation is an equivalence relation. $B = \{2, 4, 6, 8\}$, Relation on C is given as $R = \{(x, y) : x, y \in B, x | y\}$

$(2, 2), (4, 4), (6, 6), (8, 8)$

$2 | 4, 2 | 6, 2 | 8$

Functions

- ▶ A Relation from set A to set B is called a function, if every element of set A has a unique image in B
- ▶ If f is a function from A to B, we write $f: A \rightarrow B$
- ▶ f is a function from A to B, $(a, b) \in f$, then $f(a) = b$ and we call b as the image of a and a as the pre-image of b .

$$f(x) = x^2, \quad \begin{matrix} \downarrow & \downarrow \\ (-3, 9) & (3, 9) \end{matrix} \quad \begin{matrix} \text{Pre-image} \\ \{-3, 3\} \end{matrix}$$

Exercise 12: Define relation R on $\mathbb{N}$ such that $R = \{(x, y) : \boxed{y = 2x}, x, y \in \mathbb{N}\}$. What is the domain, codomain and range of R ? Is this relation a function?

$\mathbb{N}$ (set of even numbers) 3 = function

Exercise 13: Examine each of the following relations given below and state in each case, giving reasons whether it is a function or not?

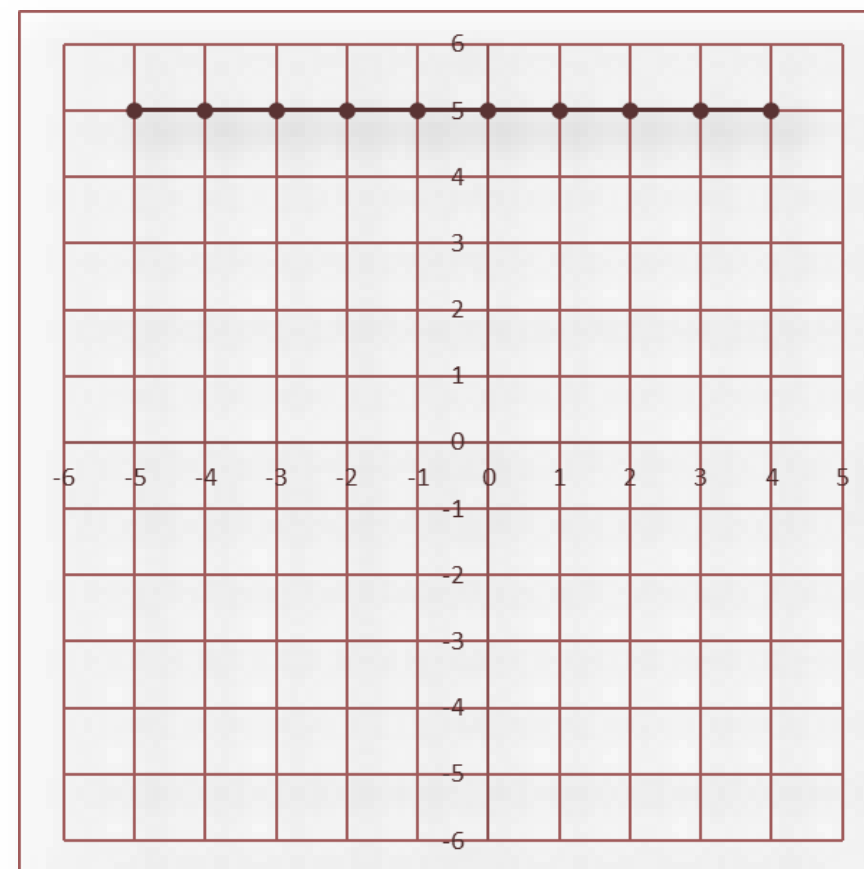
- (i) $R = \{(2, 1), (3, 1), (4, 2)\}$ $\rightarrow$ function.
- (ii) $R = \{(2, 2), (2, 4), (3, 3), (4, 4)\}$ $\rightarrow$ not a function.
- (iii) $R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7)\}$ $\rightarrow$ function.

Constant Function

- ▶ (Constant Function) $f: \underline{R} \rightarrow \underline{R}$, $y = f(x) = \underline{c}$
- ▶ Domain $\underline{= R}$, Range $= \{c\}$

$$y = f(x) = 5$$

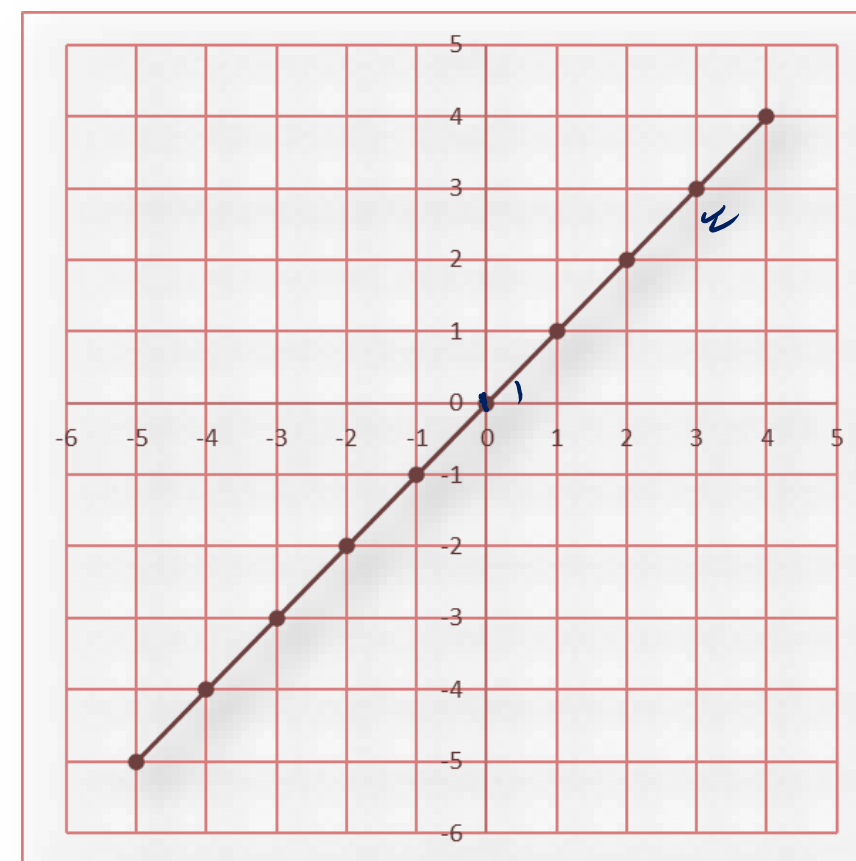
$domain = \mathbf{R}$, $range = \{5\}$



Identity Function

- ▶ (Identity Function) $f: R \rightarrow R$, $y = \underline{f(x) = x}$
- ▶ Domain = R , Range = R

$$y = f(x) = x$$
$$\text{domain} = \mathbf{R}, \text{range} = \mathbf{R}$$

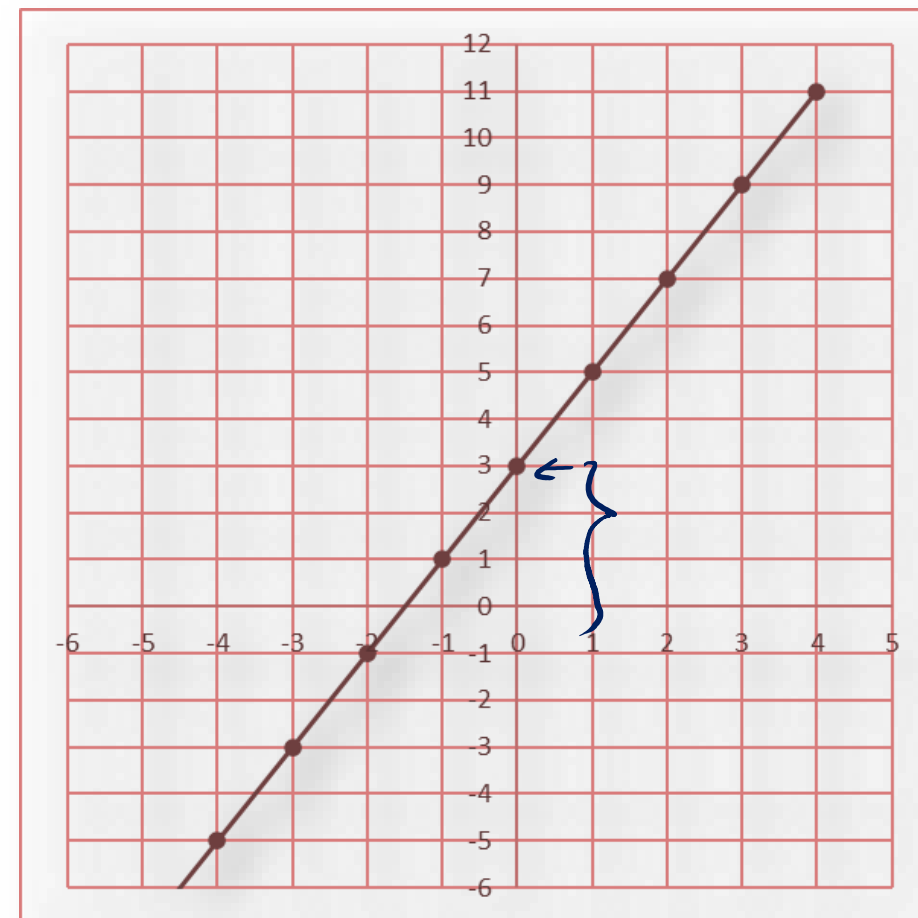


Linear Function

- ▶ (Linear Function) $f: R \rightarrow R$, $y = f(x) = \overbrace{ax + b}^{mx + c}$
- ▶ Domain = R , Range = R

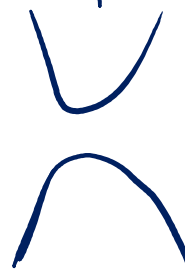
$$y = f(x) = \underline{2x + 3}$$

domain = R , range = R



Quadratic Function

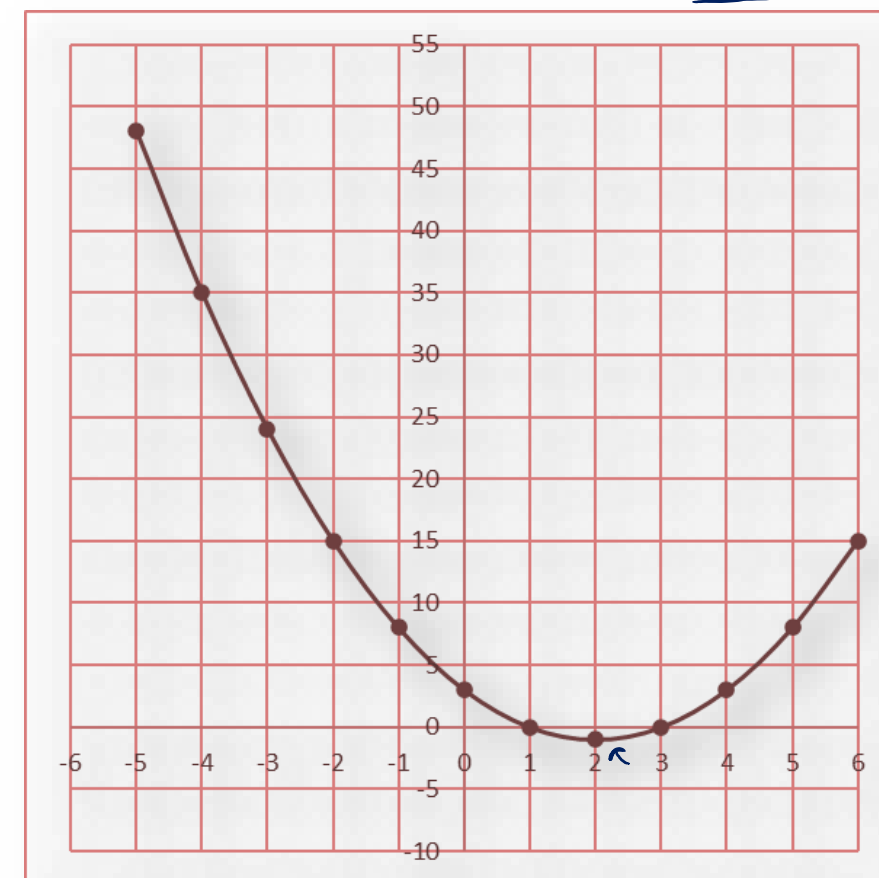
► (Quadratic Function) $f: \mathbb{R} \rightarrow \mathbb{R}$ $y = f(x) = ax^2 + bx + c$



$$y = f(x) = x^2 - 4x + 3$$

Handwritten calculation: $4 - 8 + 3 = -1$

Handwritten domain and range: $\text{domain} = \mathbb{R}, \text{range} = [-1, \infty)$



Modulus Function

► (Modulus Function) $f: \mathbb{R} \rightarrow \mathbb{R}$, $y = f(x) = |x|$

$$y = f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$$\begin{aligned} |5| &= 5 \\ |(-5)| &= 5 \\ -(-5) &= 5 \end{aligned}$$

$$\sqrt{25} = 5$$

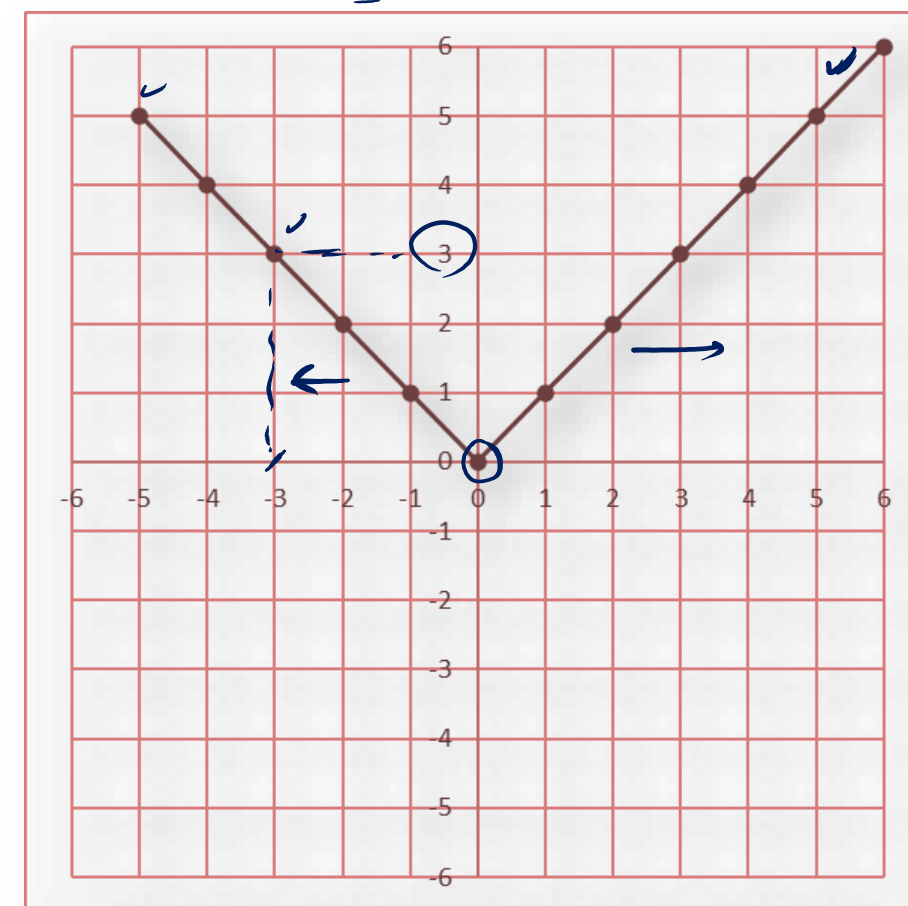
$$\sqrt{x^2} = |x| \quad \boxed{x^2 - 25 = 0}$$

$$\begin{matrix} (-x) & (+x) \\ \uparrow & \uparrow \end{matrix}$$

$$(x+5)(x-5) = 0 \quad \begin{matrix} x^2 = 25 \\ x = \sqrt{25}, \pm 5 \end{matrix}$$

$$y = f(x) = |x|$$

domain = $\mathbb{R}$, range = $[0, \infty)$ or $\mathbb{R}^+$



Indicator Function

► (Indicator Function) $f: R \rightarrow R$, $y = f(x) = 1_A(x)$

$$f(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

$$y = f(x) = (C) \cdot \underline{1_A(x)}$$

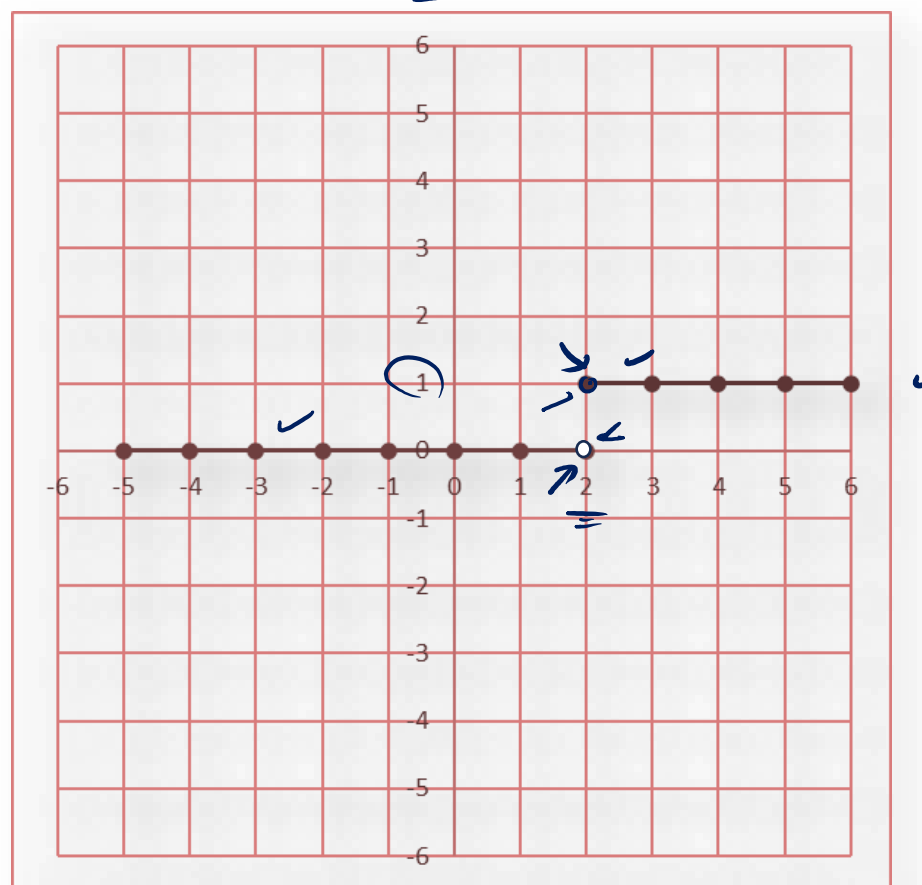
digital

$C \times 1_{S \geq K} = \text{payoff of digital option}$

$$y = f(x) = 1_{x \geq 2}(x)$$

domain = R , range = $\{0,1\}$

$x \geq 2$ ✓ $x = 2$



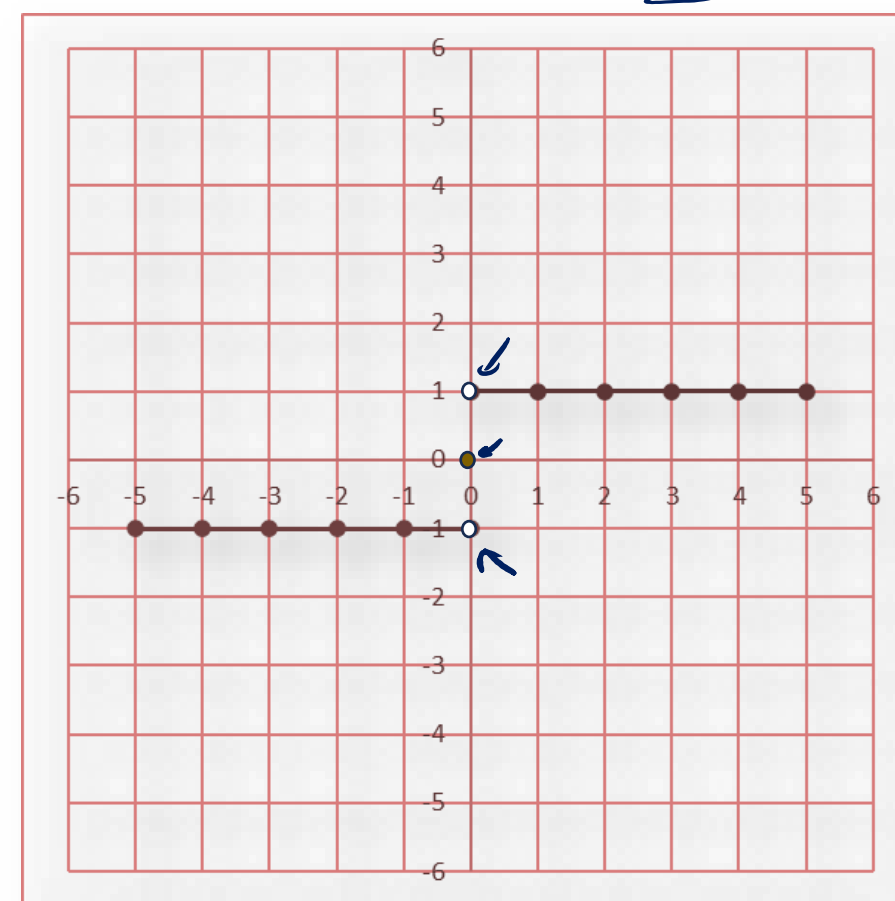
Signum Function

► (Signum Function) $f: \mathbb{R} \rightarrow \mathbb{R}$, $y = f(x) = \underline{\underline{\text{sgn}(x)}}$

$$y = f(x) = \text{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

$$y = f(x) = \text{sgn}(x)$$

$$\text{domain} = \mathbb{R}, \text{range} = \underline{\underline{\{-1, 0, 1\}}}$$



Exponential Function

► (Exponential Function) $f: R \rightarrow R$, $y = f(x) = e^x$

$$y = f(x) = e^x$$

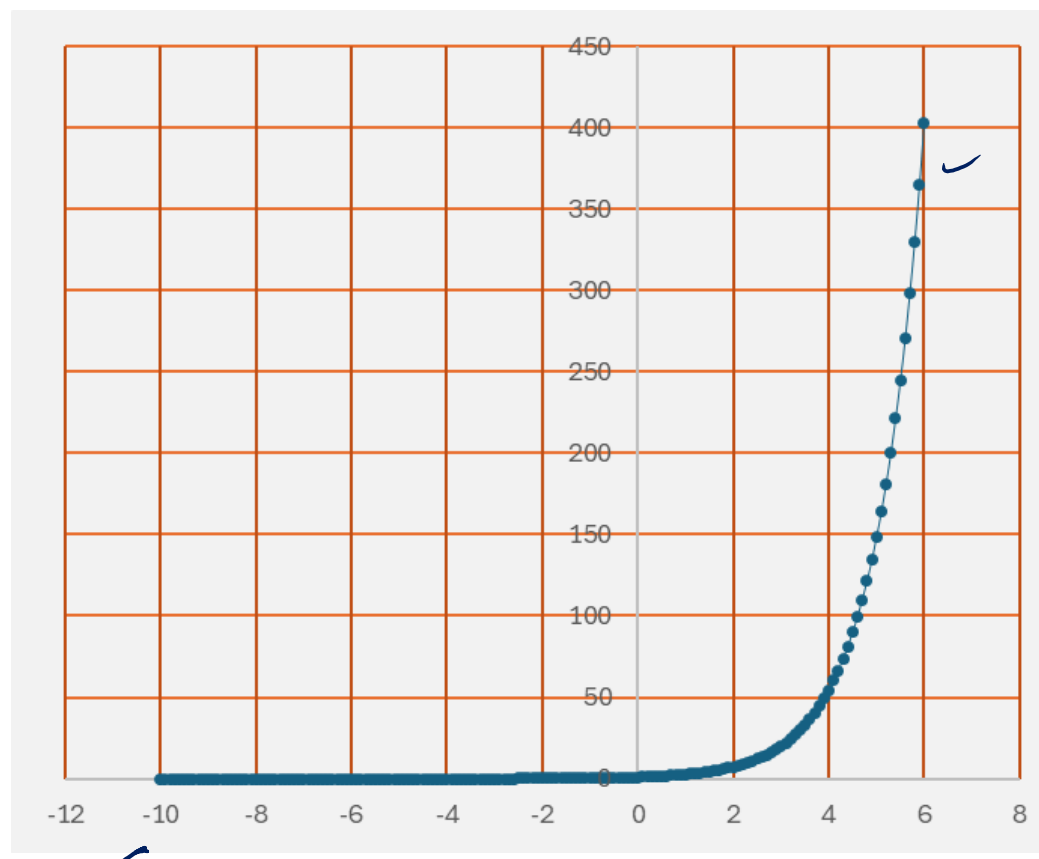
$$\text{domain} = \underline{R}, \text{range} = \underline{R^+} \quad [0, \infty)$$

$$e^{-\infty} = 0$$

$$x = ?$$

$$e^x = 1$$

$$\left(e^{\boxed{bx}} \right) \rightarrow \text{exponent}$$



Logarithmic Function

► (logarithmic Function) $f: R \rightarrow R$, $y = f(x) = \ln(x)$

$$\log_{10} x$$

$$\log_2 x$$

$$y = f(x) = \ln(x)$$

domain = R^+ , range = (R)

✓ $\ln(AB) = \ln A + \ln B$
 ✓ $\ln(A/B) = \ln A - \ln B$
 ✓ $\ln A^B = B \ln A$

$$\ln(x) = 1$$

$$\ln(x) = 0$$

$$\ln(e) = 1$$

$$\ln_e x = 0$$

$$e^0 = x = 1$$

$$\log_y x = t$$

exponential

$$y^t = x$$

$$2^3 = 8$$

$$\log_2 8 = 3$$

$$\ln(0) = ? (x) = (-\infty)$$

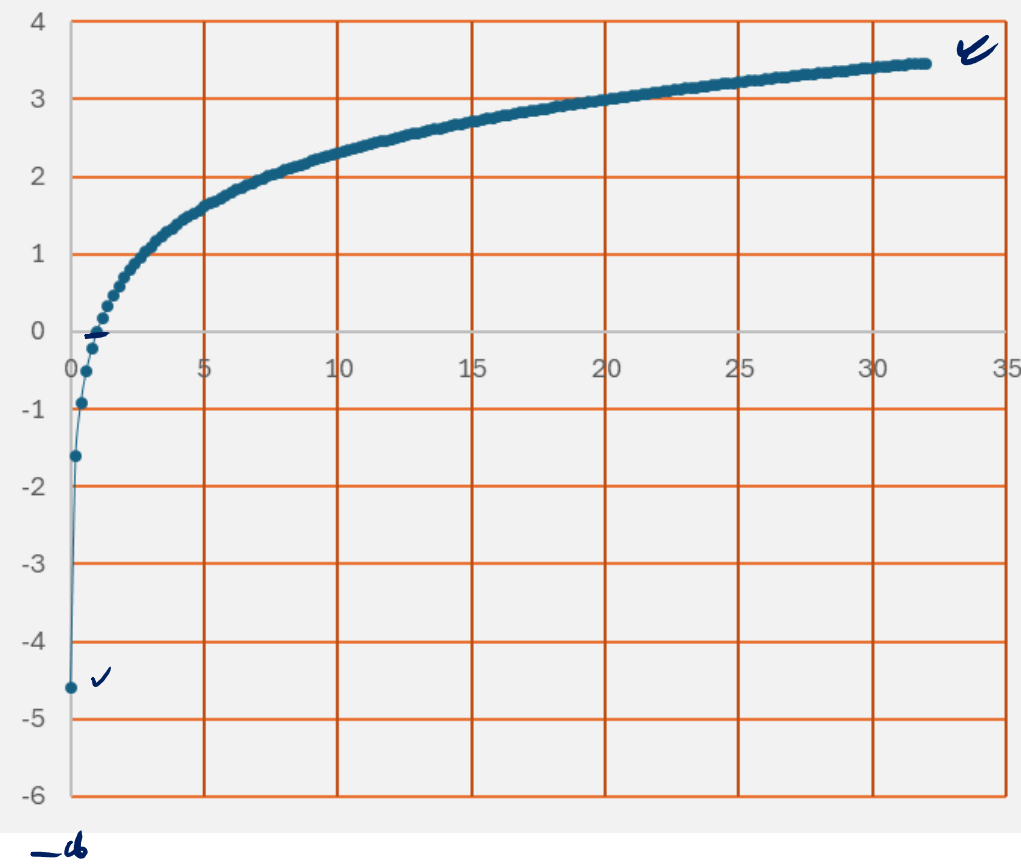
$$e^x = 0$$

$$e^{-\infty} = 0$$

$$\ln(1/2) \text{ (-ve)}$$

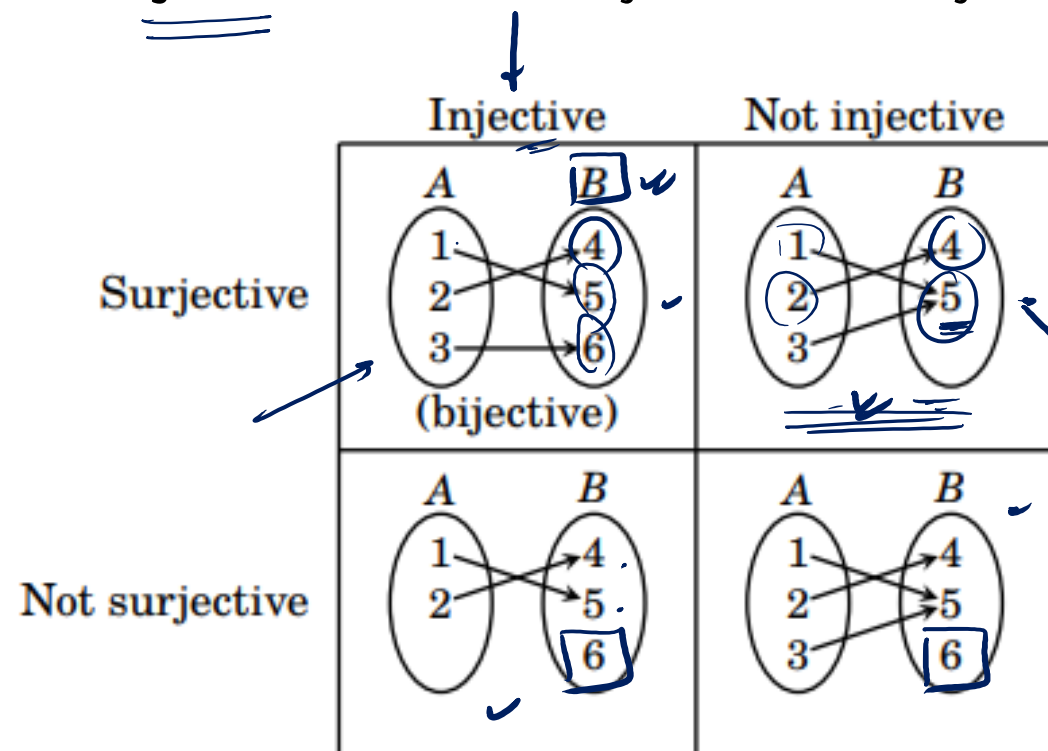
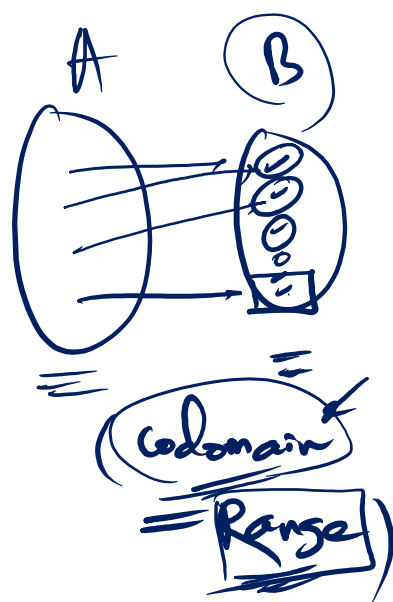
$$\ln(3/4) \text{ (-ve)}$$

$$\ln_e(-2)$$



Function Properties

- ▶ A function $f: A \rightarrow B$ is **Injective** (one-to-one) if $\forall a, b \in A, f(a) = f(b) \Rightarrow a = b$
- ▶ A function $f: A \rightarrow B$ is **Surjective** (onto) if $\forall b \in B$, there exists $a \in A$ such that $f(a) = b$. In other words, for every element in B , there is a path from A . We can also state that codomain = Range.
- ▶ A function $f: A \rightarrow B$ is **Bijjective** if it's both Injective and Surjective.



$$f(1) = f(3) = 5$$

$$1 \neq 3$$

Diagram showing a function $f: A \rightarrow B$ where A has 2 elements and B has 3 elements. The mapping is $f(1)=4, f(2)=5$. This is injective but not surjective.

$|B| > |A|$ injective

$|B| < |A|$ surjective

Function Properties

Exercise 14: Comment if the following functions are Bijective? $(\mathbb{R} \rightarrow \mathbb{R})$

(i) x^2 , (ii) x^3 -
 $\equiv y$
 $(-2)^2 = 4, 2^2 = 4 \rightarrow$ not injective $(3, 9), (-3, 9)$
 $(-2)^3 = -8, 2^3 = 8 \rightarrow$ not surjective $(, -2)$

Exercise 15: Show that the function $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x} + 1$ is injective but not surjective

$$\frac{1}{x} + 1 = \frac{1}{y} + 1 \Rightarrow x = y$$

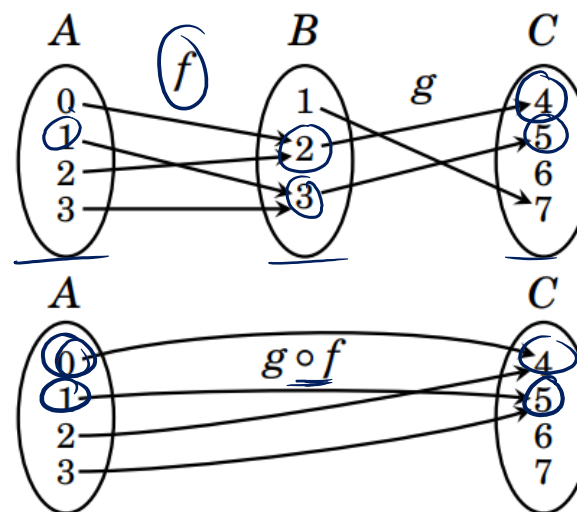
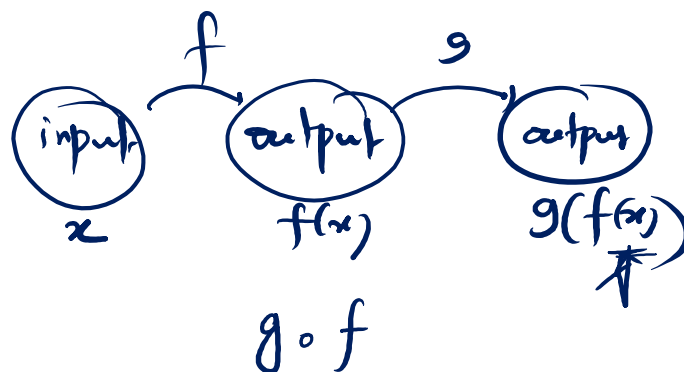
► If $|A| > |B|$, then f is not injective. ✓

► If $|A| < |B|$, then f is not surjective. ✓

Composition

- Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$ with the codomain of f being equal to the domain of g then

$$\underline{g \circ f(x)} = \underline{g(f(x))}$$



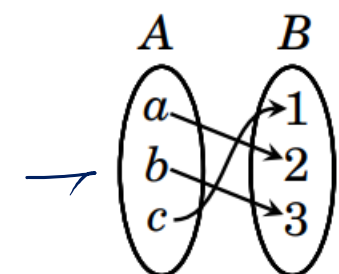
- Example : $A = \{a, b, c\}$, $B = \{0, 1\}$, $C = \{1, 2, 3\}$. Let $f: A \rightarrow B$ be the function $f = \{(a, 0), (b, 1), (c, 0)\}$, and let $g: B \rightarrow C$ be $g = \{(0, 3), (1, 1)\}$. Then $\underline{g \circ f} = \{(a, 3), (b, 1), (c, 3)\}$

Inverse

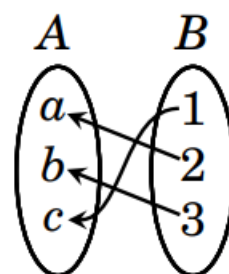
► Given the relation R from A to B , the inverse Relation of R is the relation from B to A defined as

$$R^{-1} = \{(\underline{y}, \underline{x}) : (\underline{x}, \underline{y}) \in R\}$$

Example 1 :

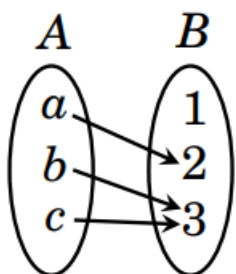


$$f = \{(\underline{a}, \underline{2}), (\underline{b}, \underline{3}), (\underline{c}, \underline{1})\}$$

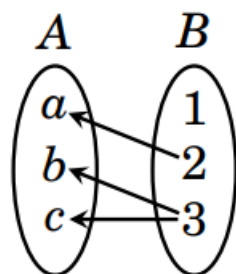


$$f^{-1} = \{(\underline{2}, \underline{a}), (\underline{3}, \underline{b}), (\underline{1}, \underline{c})\} \rightarrow \checkmark$$

Example 2 :



$$g = \{(\underline{a}, \underline{2}), (\underline{b}, \underline{3}), (\underline{c}, \underline{3})\}$$

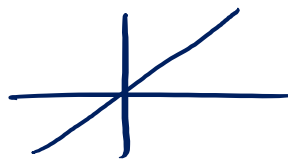


$$g^{-1} = \{(\underline{2}, \underline{a}), (\underline{3}, \underline{b}), (\underline{3}, \underline{c})\} \rightarrow \times$$

Inverse

▶ The inverse of a function exists if it's **bijjective**

▶ Also, $f^{-1} \circ f = \text{identity}$

$\nearrow \underline{\underline{f^{-1} \circ f = \text{identity}}} \searrow$
 $\boxed{f^{-1}(f(x)) = x}$


Exercise 16: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^3 + 1$ is bijective. Find its inverse. Check $f^{-1} \circ f = \text{identity}$

$$y = x^3 + 1$$

$$x = y^3 + 1 \Rightarrow y = (x - 1)^{1/3} \quad f^{-1}(x - 1)^{1/3}$$

Exercise 17: Consider this function $f : \mathbb{R} \rightarrow \mathbb{R}^+$ defined as $f(x) = e^{x^3 + 1}$. Find its inverse. Check $f^{-1} \circ f = \text{identity}$

$$f^{-1}(f(x)) = (f(x) - 1)^{1/3} = (x^3 + 1 - 1)^{1/3} = (x^3)^{1/3} = x$$

Image and Pre-Image of sets

- ▶ Suppose $f : \underline{A} \rightarrow \underline{B}$ is a function.
 - ▶ 1. If $X \subseteq \underline{A}$ the image of X is the set $f(X) = \{f(x) : x \in X\} \subseteq \underline{B}$.
 - ▶ 2. If $Y \subseteq \underline{B}$, the pre-image of Y is the set $f^{-1}(Y) = \{x \in A : f(x) \in Y\} \subseteq A$.

- ▶ Let $f : \underline{A} \rightarrow \underline{B}$ be $f = \{(s,4), (t,8), (u,8), (v,1), (w,2), (x,4), (y,6), (z,4)\}$

- ▶ f is not invertible as it's neither Injective nor Surjective, however

- ▶ Note the following :

- ▶ $f(\{s, t, u, z\}) = \{8, 4\}$
- ▶ $f(\{s, x, z\}) = \{4\}$
- ▶ $f(\phi) = \phi$
- ▶ $f^{-1}(\{4\}) = \{s, x, z\}$
- ▶ $f^{-1}(\{4, 9\}) = \{s, x, z\}$
- ▶ $f^{-1}(\{9\}) = \phi$
- ▶ $f^{-1}(4) = \text{does not exist}$

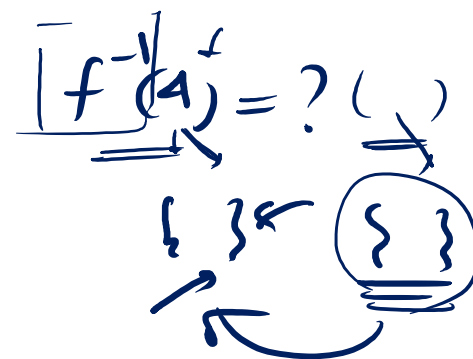
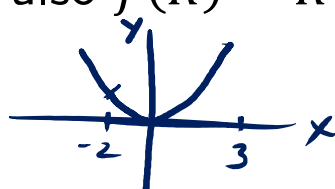


Image and Pre-Image of sets

- Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2$. Note that $f(\{0,1,2\}) = \{0,1,4\}$ and $f^{-1}(\{0,1,4\}) = \{-2,-1,0,1,2\}$. This shows that $f^{-1}(f(X)) \neq X$ in general.
- Also, note that $f([-2,3]) = [0,9]$, $f^{-1}([0,9]) = [-3,3]$, also $f(\mathbb{R}) = \mathbb{R}^+$, $f^{-1}([-2,-1]) = \phi$
- 

Exercise 18: Consider this function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2 + 3$. Find $f([-3,5])$ and $f^{-1}([12,19])$

$$f(x) = x^2 + 3$$

$$f(\{0,1,2\}) = \{0,1,4\}$$

$$f^{-1}(\{0,1,4\}) = \{-2,-1,0,1,2\}$$